MILITARY SPECIFICATION

TRACER, PROJECTILE: 11820569 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for, and the loading, assembling and packaging for one type of tracer designated as Tracer, Projectile, 11820569 Assembly for use on Projectile, 105MM, HEP-T, M393A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging: Packing & Marking for Shipment of Inert Ammunition Components
MIL-STD-1234	- Pyrotechnics: Sampling, Inspection and Testing
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Phila, PA 19137, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1315

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2.1 (cont'd)

DRAWINGS

Armament Command

C11820569	- Tracer, Projectile, 11820569 Assembly
C9321363	- Packing & Marking for Box, Fiberboard, Packing, Ammunition, For Tracer, Projectile, 11820569 Assembly
C9321365	- Packing & Marking For Box, Packing, Ammunition, For Tracer, Projectile, 11820569 Assembly
LI-11820569	- Index of Inspection Equipment Lists for Tracer, Projectile, 11820569 Assembly

2.2. Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulation

49CFR 71-90 -- Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission regulations are now a part of this Code of Federal Regulations (1967 Edition and revision) available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20315. Orders for the above publication, cite 49CFR 71-90 (latest revision).).

ANSI-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018).

ASTM Method B-117 - Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 General.

3.1.1 Tracer and igniter compositions. The materials used in the tracer and igniter compositions shall comply with the requirements specified on the drawings and referenced specifications.

3.1.2 Metal parts. The materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.1.3 Tube. The materials used in the tube shall comply with the requirements specified on the applicable drawing and referenced specification.

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3.2 Metal defects.

3.2.1 Cup.-The cup, including the cavity, shall be free from cracks, splits, cold shuts, pipe, inclusions, folds, seams and other metal defects.

3.2.2 Cup cavity.-The cup cavity shall be free of scale, fins, burrs, drawmarks, laminations, imbedded foreign matter, pits, porosity and sharp edges.

3.2.3 Visual standards.-Visual standards for metal defects (3.2.2) will be establsihed for each contract after the first month's production. (See 6.2)

3.3 Workmanship.-The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following:

3.3.1 Surface finish.-The requirements for surface finish are as detailed on the applicable drawing.

3.3.2 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurous to personnel handling the item.

3.3.3 Foreign matter.-No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter. Particular attention shall be given to the cavity of the body assembly, to insure that no grit, scale, or other foreign matter remains therein after final cleaning.

3.4 Protective coating.-Protective coating shall be in accordance with the applicable drawing and the specifications referenced thereon.

3.5 Moisture content.

3.5.1 Tracer composition.-The moisture content of the tracer composition at the loading station at the time of loading of the tracer composition shall not exceed 0.10% when tested as specified in 4.3.3.1 and 4.4.1.

3.5.2 Igniter composition.-The moisture content of the igniter composition at the loading station at the time of loading the igniter composition shall not exceed 0.14% when tested as specified in 4.3.3.1 and 4.4.1.

3.6 Tracers.-The tracers shall comply with all the requirements of Drawing C11820569 and with all the requirements specified in the applicable specifications. The tracer shall be free of dirt, chips, cracks, grease, rust, and other foreign matter. The cleaning method should not be injurous to the metal nor shall the tracer be contaminated by the cleaning agents used.

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3.7 Proving ground test.-The tracers shall ignite and there shall be visible evidence of burning of the tracer composition of 2.00 seconds minimum at ambient temperature, 125 degrees Fahrenheit (+125°F) and minus 40 degrees Fahrenheit (-40°F) when tested as specified in 4.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.1.3 Inspection criteria.

a. Successful passage of all inspections required by the drawings and specifications shall be considered as a condition precedent to performing the proving ground functional test. (See 3.7).

b. Acceptance or rejection of a lot by the Government shall be contingent on successful passage of the Proving Ground functional tests.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of assemblies and parts as follows:

55 Tracer Assemblies, complete, including protective coating
5 Cups, including protective coating
5 Cups, uncoated
5 Tubes

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4.2.1 (Cont'd)

Fifty-five of the above complete tracer assemblies shall be forwarded to a designated proving ground for burn time testing and other characteristics listed in 3.9. The samples shall be manufactured using the same materials, equipment, processes and procedures, including packaging and packing as will be used in regular production and shall be obtained from the same source of supply. A first article sample or portion thereof as directed by the procuring agency shall also be submitted whenever ther is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacture process, material used, drawing, specifications, or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Tracer assembly lot.-A tracer assembly lot shall consist of tracer assemblies produced by one manufacturer, under one contract, in one unchanged process in accordance with the same drawing, revision, same specification and same specification revision.

4.3.2 Examination.-Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4x) magnification may be used in performing visual examination. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

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4.3.2.1 Tube, Drawing A11820736, covering a detail of Drawing C11820569.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Inside diameter	Gage
202. Length	Gage
203. Wax missing	Visual

4.3.2.2 Cup, Tracer, (prior to applying compound) Drawing B11820659, covering a detail of Drawing C11820569.

CRITICAL	
None defined	
MAJOR	
101. Outside diameter, min.	Gage
102. Length, min.	Gage
MINOR	
201. Inside diameter	Gage
202. Thickness thru bottom	Gage
203. Burrs	Visual
204. Foreign matter	Visual
205. Crack, puncture or other damage	Visual

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4.3.2.3 Cup, Tracer, (prior to insertion of tube) Drawing B11820659, covering a detail of Drawing C11820569.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Coating compound missing, or not completely covering interior surface of cup	Visual
MINOR	
None defined	

4.3.2.4 Cup, Tracer (prior to loading) see Drawing B11820659.

CRITICAL	
None defined	
MAJOR	
101. Tube missing	Visual
MINOR	
None defined	

4.3.2.5 Tracer Assembly, Drawing C11820569.

CRITICAL	
None defined	
MAJOR	
101. Outside diameter, max.	Gage
102. Total length, max.	Gage
103. Charge missing or less than min, required depth from shoulder	Visual
104. Ignitor charge chipped or broken on any one step in excess of 1/3 of the total area of the step, or total aggregate of chipped or broken area of all steps in excess of 1/3 of the total area of the largest step	Visual
105. No Multiple parts if not required	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MINOR	
None defined	
4.3.2.6 Box, Fiberboard, packing, prior to sealing, Drawing C9321363	
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Any packing material missing	Visual
202. Tracer missing	Visual
4.3.2.7 Box, Fiberboard, after sealing, Drawing C9321363.	
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Sealing strip fails to completely seal carton	Visual
202. Carton torn, cut or punctured	Visual
203. Marking missing, misleading or unidentifiable	Visual
4.3.2.8 Packing & marking, for Box, Packing, Ammunition, Drawing C9321365.	
CRITICAL	
None defined	
MAJOR	
101. Box damaged exposing contents	Visual
102. Board loose or protruding nail	Visual
103. Strapping broken, missing or loose	Visual

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
104. ICC Nomenclature missing, misleading or unidentifiable	Visual
105. Cover improperly assembled	Visual
106. Contents loose	Manual
MINOR	
201. Strapping improperly engaged	Visual/Manual
202. Marking missing, misleading, or identifiable	Visual

4.3.3 Testing.

4.3.3.1 Moisture and total volatile content of tracer composition and igniter composition.-The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. The contractor shall test a minimum of one randomly selected sampe from each batch or from each 8 hours production, whichever is the leaser amount. Composite samples shall not be used for determining compliance. Failure to comply with the applicable requirement shall be cause for rejection of the parent quantity represented by the sample. Testing shall be performed as specified in 4.4.1.

4.3.3.2 Proving ground.

4.3.3.2.1 Functioning (see 3.7).-This test shall be performed as specified in 4.4.3.1.

4.3.3.2.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter I and AQL of 1.50 percent. If any tracer fails to ignite or fails to product visible evidence of burning, the lot shall be rejected.

4.3.3.2.1.2 After three consecutive lots have met the criteria of 4.3.3.2.1.1, the sampling plan and procedure for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter F and an AQL of 1.50 percent. If any tracer fails to ignite or produce visible evidnece of burning, the lot shall be rejected.

4.3.3.2.1.3 Refiring.-If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional tracers as required.

4.3.4 Inspection equipment.-LI-11820569 identifies the inspection equipment required to perform the examinations and tests prescribed herein.

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4.4 Test methods and procedures.

4.4.1 Determination of moisture and total volatile content of tracer composition and igniter composition. The moisture and total volatile content of the tracer composition and igniter composition shall be determined in accordance with MIL-STD-1234, Method 102.2. (See 6.4)

4.4.2 Protective coating. The procedure for testing the cleaning of components and the application of protective coatings shall be in accordance with the criteria for testing, as specified in the applicable specification. The salt spray test shall be performed utilizing the procedures and equipment proscribed by ASTM Method B117.

4.4.3 Proving ground.

4.4.3.1 Ballistic tracer functioning (acceptance test). The sample tracer shall be assembled to inert or HE loaded 105MM, HEP-T, M393A2 cartridges. The assembled rounds shall be fired at the temperatures listed in Table I from the appropriate weapon and the tracer burn times recorded. A timing device capable of measuring time to the nearest 0.01 seconds shall be used to measure burning time. Suitable aids, such as binoculars, should be used if necessary to insure visible contact with the tracer (see 6.9 and 6.10).

TABLE I

Temperature	Quantity
Ambient	25
+125°F 1/	15 2/
- 40°F 1/	15 2/

1/ The rounds shall be temperature conditioned for a minimum period of 18 hours.

2/ After first article sample and three (3) consecutive production lots have been tested successfully, the temperature conditioning phase may be discontinued with the approval of the procuring agency. However, implementation of the temperature conditioning phase shall be instituted whenever directed by the procuring contracting officer (P.C.O.).

4.4.4 Government use of contractor's inspection equipment. The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 The tracer assemblies shall be preserved and packaged in accordance with drawing C9321363 and the requirements of Level C in MIL-STD-1169.

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5.2 Packing.

5.2.1 The cartons shall be packed in accordnace with drawing C9321265 and the requirements of Level C in MIL-STD-1169.

5.3 Marking.

5.3.1 The cartons and boxes shall be marked in accordance with drawings C9321363, C9321265 and MIL-STD-1168.

5.4 Data cards. Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, this characteristic amy be measured or gaged to determine acceptability.

6.2 Visual standards. Visual standards to be established at contractor's plant in accordance with 3.2.3.

6.3 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.4 Moisture. The igniter and tracer compositions, and the pellets formed thereof, are hygroscopic and will deteriorate rapidly when exposed to moisture. Small amounts of moisture present in or absorbed by the compositions or pellets, prior to or after loading into tracers will eventually result in unsatisfactory functioning. It is highly desireable that the ingredients, compositions, pellets, and loaded tracer be carefully safeguarded from exposure to moisture and humid atmospheres. For assurance of a satisfactory loaded tracer, it is recommended that the blending and pelletizing buildings be air conditioned in a manner which will hold humidity of the atmosphere as low as practicable.

6.5 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for approval of the responsible technical agency that their effectivness is equal to or better than the quality assurance provisions specified herein.

6.6 When a product is being inspected in accordance with MIL-STD-1235 and a defective characteristic is found, the product produced prior to the discovered defect will be screened back 2000 tracer assemblies and all defecte culled out.

6.7 Shipping. It is not intended to ship these compositions or pellets. It is intended that the compositions and pellets shereof be prepared at the plant where the loading of the tracers will be accomplished. If shipment is necessary, Department of Transportation (DOT) Class and Marking cited on the affected item

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6.7 (Cont'd)

drawing is applicable. Material must be adequately and properly packaged in accordance with applicable DOT Packaging Requirements (Para 173.61, CODE OF FEDERAL REGULATIONS (CFR) Title 49.).

6.8 Hazard notice. The tracer described herein and certain of its components are flammable and/or explosive and consequently present hazards in manufacture, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.9 Combining of proving ground tests. When the contractor for the tracer is also contractor for the complete round or one of the loaded components thereof, the proving ground tests may be combined in order to save expense, upon agreement between the procuring activity and the contractor.

6.10 Submission of firing and test data. In addition to the normal distribution of records for each lot of tracers, when the item is procured by the Department of Defense, copies should be sent to:

Commander
Frankford Arsenal
ATTN: SARFA-QAA-Q(1) (1 copy)
SARFA-MDY-P (1 copy)
Philadelphia, PA 19137

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1315-A292

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